

NITHISH P

Software Developer | Backend Systems | Industrial Automation | MES Integration

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PROFESSIONAL SUMMARY

Software Developer with 1+ year of experience designing and delivering high-performance backend systems, real-time data pipelines, and industrial automation solutions. Specialized in Go-based MQTT/OPC UA adapters, AVEVA MES ecosystem integration (MES, WorkTask, System Platform, Historian, Connect, Enterprise Integration), and full-stack web applications using Node.js and React.js. Proven track record of replacing legacy message brokers, building custom TCP/MQTT brokers, and engineering scalable workflows for mission-critical manufacturing environments. Adapt at collaborating on architecture design, performance tuning, and MCP server development.

TECHNICAL SKILLS

Languages: Go, Node.js (JavaScript/TypeScript), Python, Java, C#, VB.NET, C, PowerShell

Frontend: React.js, ReactFlow, HTML5, CSS3, JavaScript, TypeScript, OMI ViewApps

Messaging & Protocols: MQTT (QoS 0/1), NATS JetStream, TCP/IP, OPC UA, REST APIs, RabbitMQ

Databases: Microsoft SQL Server (MSSQL), MySQL, PostgreSQL

AVEVA / Industrial: AVEVA MES, WorkTask, System Platform, Historian, Connect, Enterprise Integration (EI), OMI ViewApps

Frameworks & Tools: .NET, Spring Boot, Vite, Git, Postman, VS Code, Eclipse, npm

AI & Integration: MCP Server Development, Claude API (Anthropic), Multi-source data integration

PROFESSIONAL EXPERIENCE

Software Developer | Ficus IT Service Private Limited

Erode, Tamil Nadu | May 2025 – Present

AVEVA MES & Industrial Integration

- Designed and developed a custom MQTT Adapter in Go that ingests data from MQTT protocol and publishes directly to AVEVA Connect, eliminating the RabbitMQ broker dependency and reducing infrastructure overhead.
- Built an OPC UA Adapter in Go to receive and process large-scale industrial tag data from plant floor systems, integrating with AVEVA System Platform and Historian.
- Developed an ERP-to-MES integration DLL using .NET (C#) to connect enterprise resource planning systems with AVEVA Enterprise Integration (EI), enabling seamless bidirectional data exchange.
- Engineered Cloud MES modules for Job Management and Work Order (WO) Management using Node.js, React.js, and MSSQL, deployed for AVEVA MES workflows.
- Built custom label widgets and AVEVA WorkTask custom widgets in JavaScript to enhance operational UI and streamline shop-floor task management.

Backend Systems & Real-Time Data Pipelines

- Built scalable MQTT pipelines in Go subscribing to and processing 100,000+ individual topics with low latency, supporting high-throughput industrial data streams.
- Implemented custom TCP and MQTT brokers supporting QoS 0 and QoS 1, optimized for concurrency and throughput using Go goroutines and channel-based concurrency patterns.
- Engineered batch-based message publishing to NATS JetStream, reducing CPU usage and improving system reliability under peak load conditions.
- Developed real-time data persistence services to store high-volume streaming data into Microsoft SQL Server with optimized write performance.
- Conducted Archive DB testing to validate data integrity and recovery mechanisms for mission-critical industrial data stores.
- Optimized system performance using buffer tuning, memory reuse, and concurrency control, improving throughput and stability across all services.

Workflow & AI Integration

- Developed Workflow Activity backend and frontend using Node.js and ReactFlow library, enabling configurable process flow design, execution, and error handling for manufacturing operations.
- Created an MCP (Model Context Protocol) Server using Node.js, integrating multiple data sources including AVEVA MES DB and AVEVA Historian with the Claude API (Anthropic), enabling AI-assisted querying of industrial data.
- Built React-based and OMI ViewApp dashboards for workflow visualization, system monitoring, and operational control.
- Participated in architecture design, technology evaluation, and performance tuning for mission-critical industrial systems.

Python Full Stack Development Intern | Techvolt Software Pvt Ltd

Coimbatore, Tamil Nadu | Dec 2024 – Mar 2025

- Developed full-stack web applications using Python and modern frameworks, gaining hands-on experience in backend API development and frontend integration.

KEY PROJECTS

High-Volume MQTT Streaming System — Go, MQTT, Aveva Connect, MSSQL

- Real-time Go services processing 100,000+ MQTT topics with custom broker, batch publishing to Aveva Connect, and optimized SQL Server persistence.

AVEVA MCP Server — Node.js, Claude API, AVEVA Historian, MSSQL

- Built an MCP server integrating AVEVA MESDB and Historian data sources with the Claude AI API, enabling natural language querying of industrial operational data.

Cloud MES Job & WO Management Module — Node.js, React.js, MSSQL

- Full-stack MES module for managing manufacturing jobs and work orders, with REST API backend and interactive React.js frontend.

ERP-to-MES Enterprise Integration DLL — C#, .NET, AVEVA EI

- Developed a .NET DLL for bidirectional ERP and AVEVA MES data synchronization using AVEVA Enterprise Integration middleware.

Agricultural Guidance Platform — Python CGI, MySQL

- Multi-role login system with crop advisory features, built using Python CGI and MySQL backend.

Shopping Mall Management System — Spring Boot, Angular, REST APIs

- RESTful backend with Spring Boot and Angular frontend for managing retail operations including tenants, inventory, and billing.

EDUCATION

B.E. Computer Science and Engineering (Hons – AI & ML)

Excel Engineering College, Namakkal | 2021 – 2025

CERTIFICATIONS & COURSES

- Java Full Stack Development – TNSIF (6-month program, 2024–2025)